



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Curriculum for Undergraduate Degree (B.Tech.) in Mechanical Engineering (w.e.f. AY: 2020-21)

Part I: Introduction, Theme & Category wise Credit Distribution

A. Definition of Credit:

Sl. No.	Description	Credit
1	1 Hr. Lecture (L) Per Week	1.0
2	1 Hr. Tutorial (T) Per Week	1.0
3	1 Hr. Practical/ Lab (P) Per Week	0.5
4	2 Hrs. Practical/ Lab (P) Per Week	1.0

B. Range of Credits:

As per AICTE, a student covering 160 credits during 4 years of studies as per curriculum of the Institute will be eligible to get Under Graduate B.Tech. degree. A student will be eligible to get B.Tech. degree with Honours, if he/ she completes an additional 20 credits. These could be acquired through MOOCs prescribed by the Institute. Every student admitted to the 4 years B.Tech programme is required to earn a minimum of 100 Activity Points under Mandatory Additional Requirement (MAR), in addition to the required academic grades, for getting B.Tech./ B.Tech. (Honours) degree.

C. Category wise Credit Distribution:

Sl. No.	Category	Credit Allotted	Credit as per AICTE
1	Humanities and Social Sciences including Management Courses	13	12
2	Basic Science Courses	26	25
3	Engineering Science Courses including Workshop, Drawing, Basics of Electrical/ Mechanical/ Computer etc.	24	24
4	Professional Core Courses	57.5	48
5	Professional Elective Courses relevant to chosen specialization/ branch	18	18
6	Open Elective Courses from other technical and / or emerging subjects	9	18
7	Project Work, Seminar and Internship in Industry or elsewhere	12.5	15
8	Mandatory Courses [Environmental Sciences, Induction Training, Indian Constitution, Essence of Indian Traditional Knowledge, Aptitude Skill]	Non-Credit	Non-Credit
Total		160	160



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D. Course Code and Definition:

Sl. No.	Course Code	Definitions
1	L	Lecture
2	T	Tutorial
3	P	Practical
4	BS	Basic Science Courses
5	ES	Engineering Science Courses
6	HM	Humanities and Social Sciences including Management Courses
7	PC	Professional Core Courses
8	PE	Professional Elective Courses
9	OE	Open Elective Courses
10	MC	Mandatory Courses
11	PW	Project/ Internships/ Sessional

E. Courses in different Category:

Humanities and Social Sciences including Management Courses							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	II	HM-HU 201	English	2	0	0	2
2	V	HM-HU 503	Effective Technical Communication	3	0	0	3
3	VI	HM-HU 603	Operations Research	4	0	0	4
4	VII	HM-HU 701	Economics for Engineers	3	0	0	3
Total Theory				12	0	0	12
Practical/ Sessional							
1	I	HM-HU 191	Language Laboratory	0	0	2	1
2	VI	HM-HU 691	Soft skill Development Lab	0	0	2	1
Total Practical/ Sessional				0	0	4	2
Total				12	0	4	14

Basic Science Courses							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	BS-CH 101	Chemistry	3	1	0	4
2	I	BS-M 101	Mathematics- I	3	1	0	4
3	II	BS-PH 201	Physics	3	1	0	4
4	II	BS-M 201	Mathematics - II	3	1	0	4
5	III	BS-M 303	Mathematics- III	2	1	0	3



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6	III	BS-BIO 301	Biology	2	0	0	2
7	IV	BS- M 404	Numerical methods	2	0	0	2
Total Theory				18	5	0	23
Practical/ Sessional							
1	I	BS-CH 191	Chemistry Laboratory	0	0	3	1.5
2	II	BS-PH 291	Physics Laboratory	0	0	3	1.5
Total Practical/ Sessional				0	0	6	3
Total				18	5	6	26

Engineering Science Courses including Workshop, Drawing, Basics of Electrical/ Mechanical/ Computer etc.							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	ES-CS 101	Programming for Problem Solving	3	0	0	3
2	II	ES-EE 201	Basic Electrical & Electronics Engineering	4	0	0	4
3	III	ES-ME 301	Materials Engineering	3	0	0	3
4	III	ES-ME 302	Engineering Mechanics	3	1	0	4
Total Theory				13	1	0	14
Practical/ Sessional							
1	I	ES-CS 191	Programming for Problem Solving Lab	0	0	4	2
2	I	ES-ME 192	Workshop/ Manufacturing Practices	1	0	4	3
3	II	ES-EE 291	Basic Electrical & Electronics Engineering Lab	0	0	4	2
4	II	ES-ME 291	Engineering Graphics & Design	1	0	4	3
Total Practical/ Sessional				2	0	16	10
Total				15	1	16	24

Professional Core Courses							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	III	PC-ME 301	Thermodynamics	3	1	0	4
2	III	PC-ME 302	Manufacturing Processes	4	0	0	4
3	IV	PC-ME 401	Applied Thermodynamics	3	1	0	4
4	IV	PC-ME 402	Fluid Mechanics & Fluid Machines	3	1	0	4
5	IV	PC-ME 403	Strength of Materials	3	1	0	4
6	IV	PC-ME 404	Metrology and Instrumentation	3	1	0	4



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7	V	PC-ME 501	Heat Transfer	3	1	0	4
8	V	PC-ME 502	Solid Mechanics	3	1	0	4
9	V	PC-ME 503	Kinematics & Theory of Machines	3	1	0	4
10	VI	PC-ME 601	Manufacturing Technology	4	0	0	4
11	VI	PC-ME 602	Design of Machine Elements	3	1	0	4
12	VII	PC-ME 701	Advanced Manufacturing Technology	3	0	0	3
Total Theory				38	9	0	47
Practical/ Sessional							
1	III	PC-ME 391	Practice of Manufacturing Processes	0	0	3	1.5
2	IV	PC-ME 491	Metrology and Measurement Laboratory	0	0	3	1.5
3	IV	PC-ME 492	Machine Drawing- I	0	0	3	1.5
4	V	PC-ME 591	Mechanical Engineering Laboratory I (Thermal)	0	0	3	1.5
5	V	PC-ME 592	Machine Drawing-II	0	0	3	1.5
6	VI	PC-ME 691	Mechanical Engineering Laboratory II (Design)	0	0	3	1.5
7	VII	PC-ME 791	Mechanical Engineering Laboratory III (Manufacturing)	0	0	3	1.5
Total Practical/ Sessional				0	0	21	10.5
Total				38	9	25	57.5

Professional Elective Courses relevant to chosen specialization/ branch							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	VI	PE-ME 601	Elective-I	3	0	0	3
2	VI	PE-ME 602	Elective-II	3	0	0	3
3	VII	PE-ME 701	Elective III	3	0	0	3
4	VII	PE-ME 702	Elective-IV	3	0	0	3
5	VIII	PE-ME 801	Elective V	3	0	0	3
6	VIII	PE-ME 802	Elective VI	3	0	0	3
Total Theory				18	0	0	18
Practical/ Sessional							
1							
Total Practical/ Sessional							
Total				18	0	0	18



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Open Elective Courses from other technical and / or emerging subjects							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	VII	OE-AUE 701	Open Elective- I	3	0	0	3
2	VIII	OE-AUE 801	Open Elective-II	3	0	0	3
3	VIII	OE-AUE 801	Open Elective- III	3	0	0	3
Total Theory				9	0	0	9
Practical/ Sessional							
1							
Total Practical/ Sessional							
Total				9	0	0	9

Project Work, Seminar and Internship in Industry or elsewhere							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	V	PW-ME 581	Project-I (30 hrs. Total)	0	0	2	1
2	VI	PW-ME 681	Project-II (90 hrs. Total)	0	0	4	2
3	VII	PW-ME 781	Project-III	0	0	6	3
4	VIII	PW-ME 881	Project-IV	0	0	8	4
Total Theory				0	0	20	10
Practical/ Sessional							
1	VIII	PW-ME 882	Comprehensive viva	0	0	0	1.5
Total Practical/ Sessional							
Total				0	0	20	11.5



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Mandatory Courses [Environmental Sciences, Induction Training, Indian Constitution, Essence of Indian Traditional Knowledge, Aptitude Skill]							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	IV	MC 471	Environmental Science	2	0	0	0
2	V	MC 571	Aptitude Skill Development-I	2	0	0	0
3	VI	MC 671	Aptitude Skill Development-II	2	0	0	0
4	VI	MC 672	Constitution of India	2	0	0	0
Total Theory				10	0	0	0
Practical/ Sessional							
1							
Total Practical/ Sessional							
Total				0	0	0	0